

**International Software Engineering Program
Department of Computer Engineering
School of Engineering
King Mongkut's Institute of Technology Ladkrabang**

13016223 Artificial Intelligence
Midterm Examination
30th September 2021, 13:30 – 15:30 Hrs

Instruction

Study the problem description below and answer the two questions.

Problem description:

Intelligent Cleaning Robot Vacuum Cleaner (ICRVC) is a robot for floor cleaning. When it operates, it keeps moving forward, while sucking the dust on the way it is moving over. The robot keeps the dust in its dust bag to be thrown away later by the user when the bag is full. While moving forward it can turn left/right by itself to avoid any obstacle in front of it or avoid hitting with the wall. The robot operates with a rechargeable battery, when the battery power is getting low it can find the charging station by itself using wireless communication hand-check with the station.

The robot can determine by itself whether the whole floor is now completely clean and it does not repeat cleaning the same area it has already cleaned.



Figure of an Intelligent Cleaning Robot Vacuum Cleaner

Question 1 (10 points)

Please analyze the ICRVC agent in terms of PEAS.

Question 2 (10 points)

What would be an agent architecture of the ICRVC? Please explain the detail of that architecture together with a diagram of that architecture.
